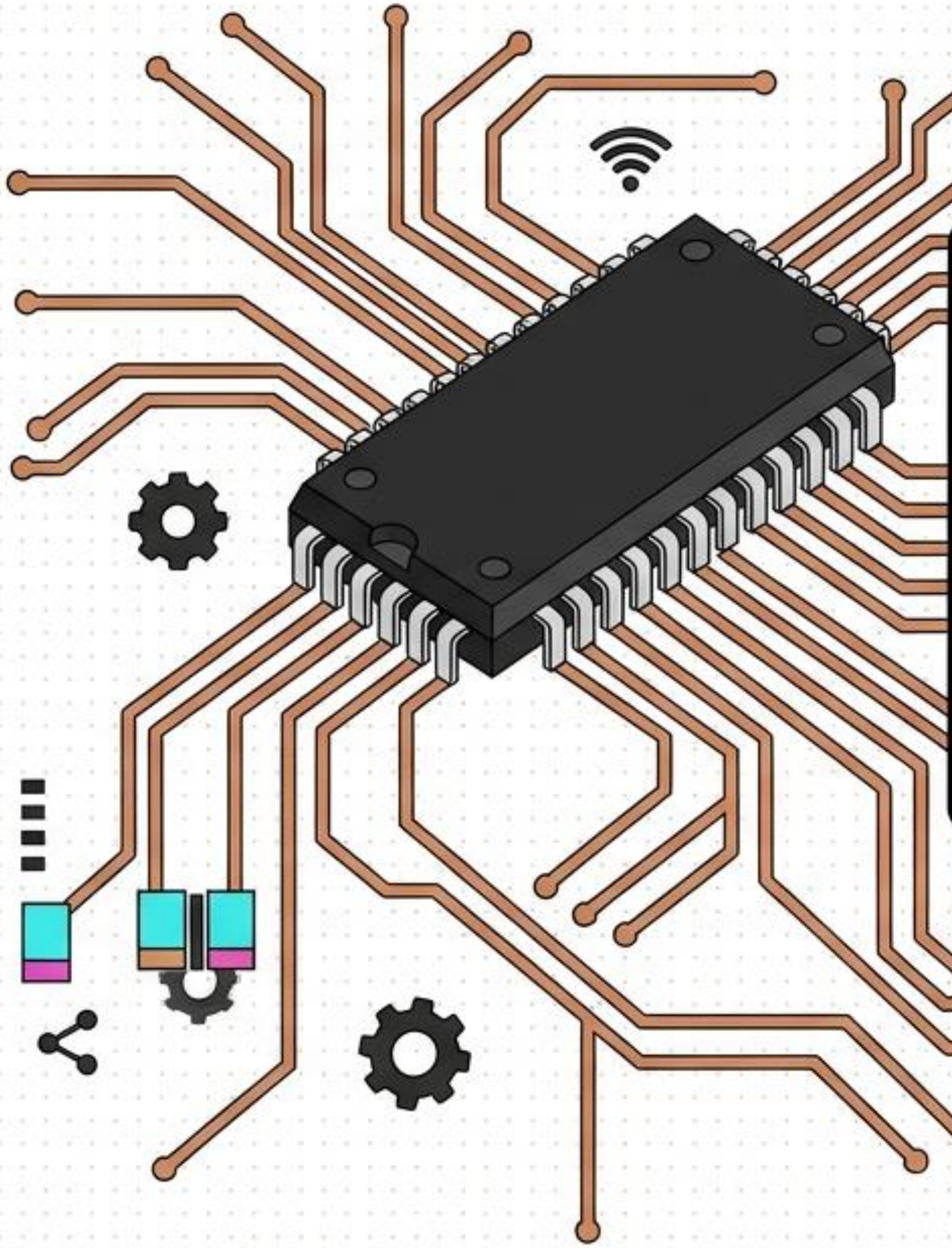


Pico IoT Adventure

Build your own automated, smart-connected world.

Choose Your Mission

🛌 Theme 1: The Smart Bedroom

- **Ambient Lighting:** Automated ambient nightlight
- **Motorized Tech:** Morning sun curtains
- **Security:** Intruder security alarm

Same Code. Different Missions.

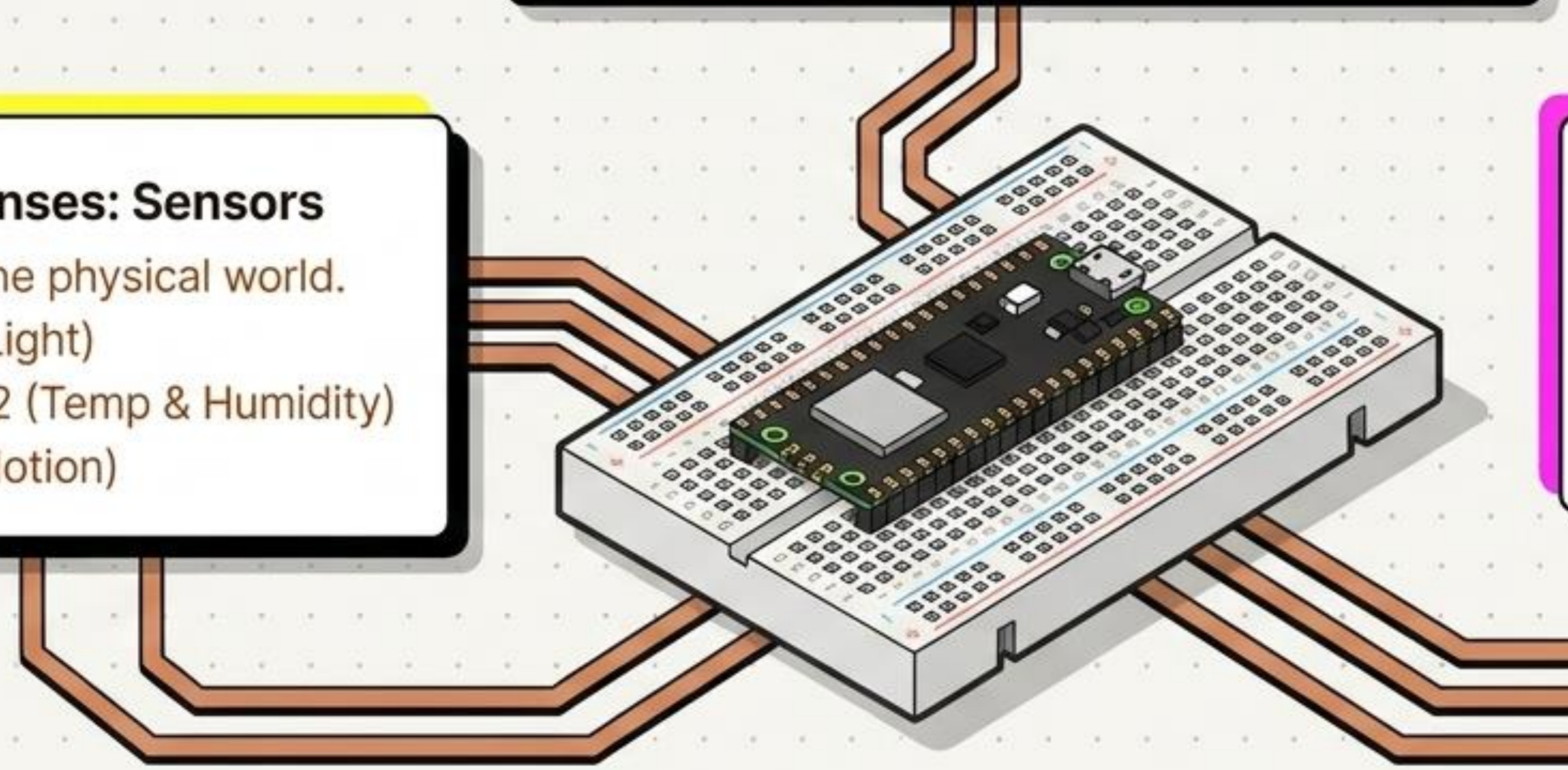
🌾 Theme 2: The Mini Smart Farm

- **Ambient Lighting:** Automated greenhouse grow light
- **Motorized Tech:** Smart sunshade deployer
- **Security:** Crop security & environment monitor

Anatomy of Your Cyborg

🧠 The Brain: Raspberry Pi Pico W

Processes logic and connects to Wi-Fi.



👁️ The Senses: Sensors

Reads the physical world.

- LDR (Light)
- DHT22 (Temp & Humidity)
- PIR (Motion)

💪 The Muscles: Actuators

Changes the physical world.

- LED (Light)
- SG90 Micro Servo (Movement)

Your Digital Toolbelt

🔧 Thonny IDE (The Wrench)

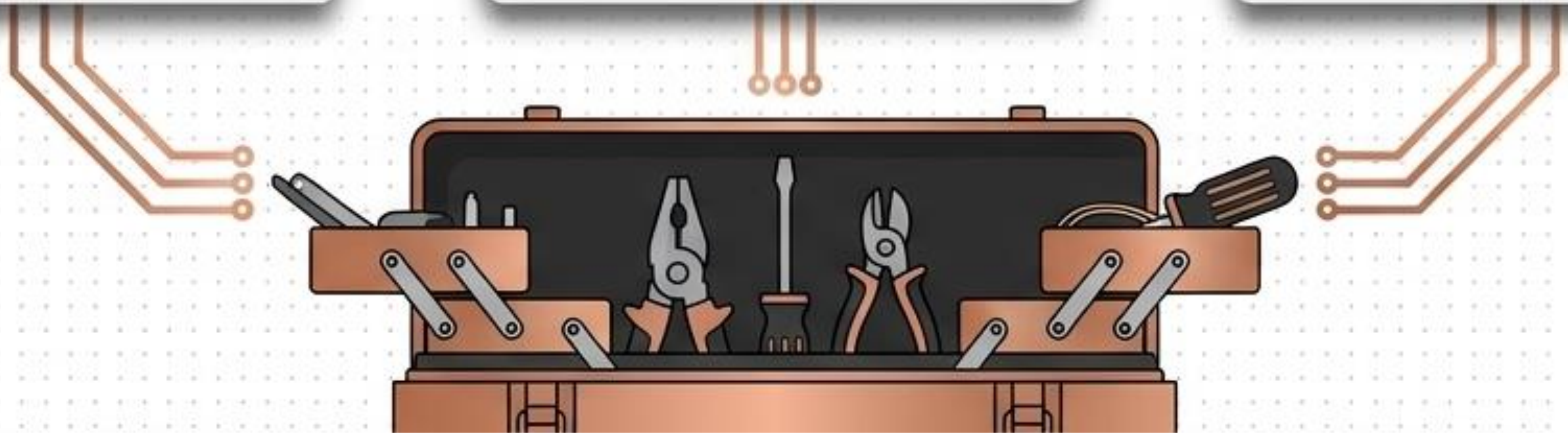
Your beginner-friendly code editor. Use it to build code and send it to the Pico.

🧠 CircuitPython (The Language)

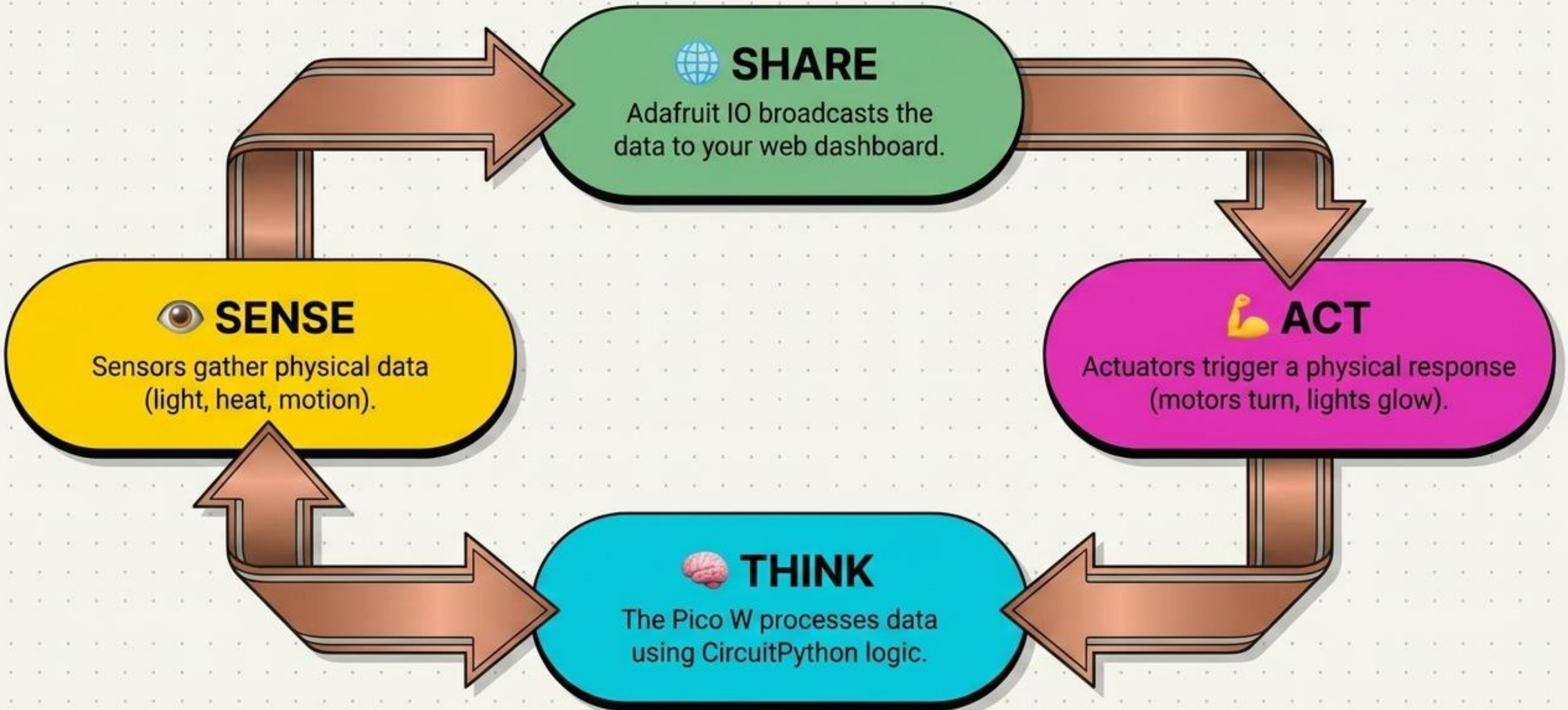
The firmware installed on the hardware that teaches your brain how to think.

☁️ Adafruit IO (The Cloud)

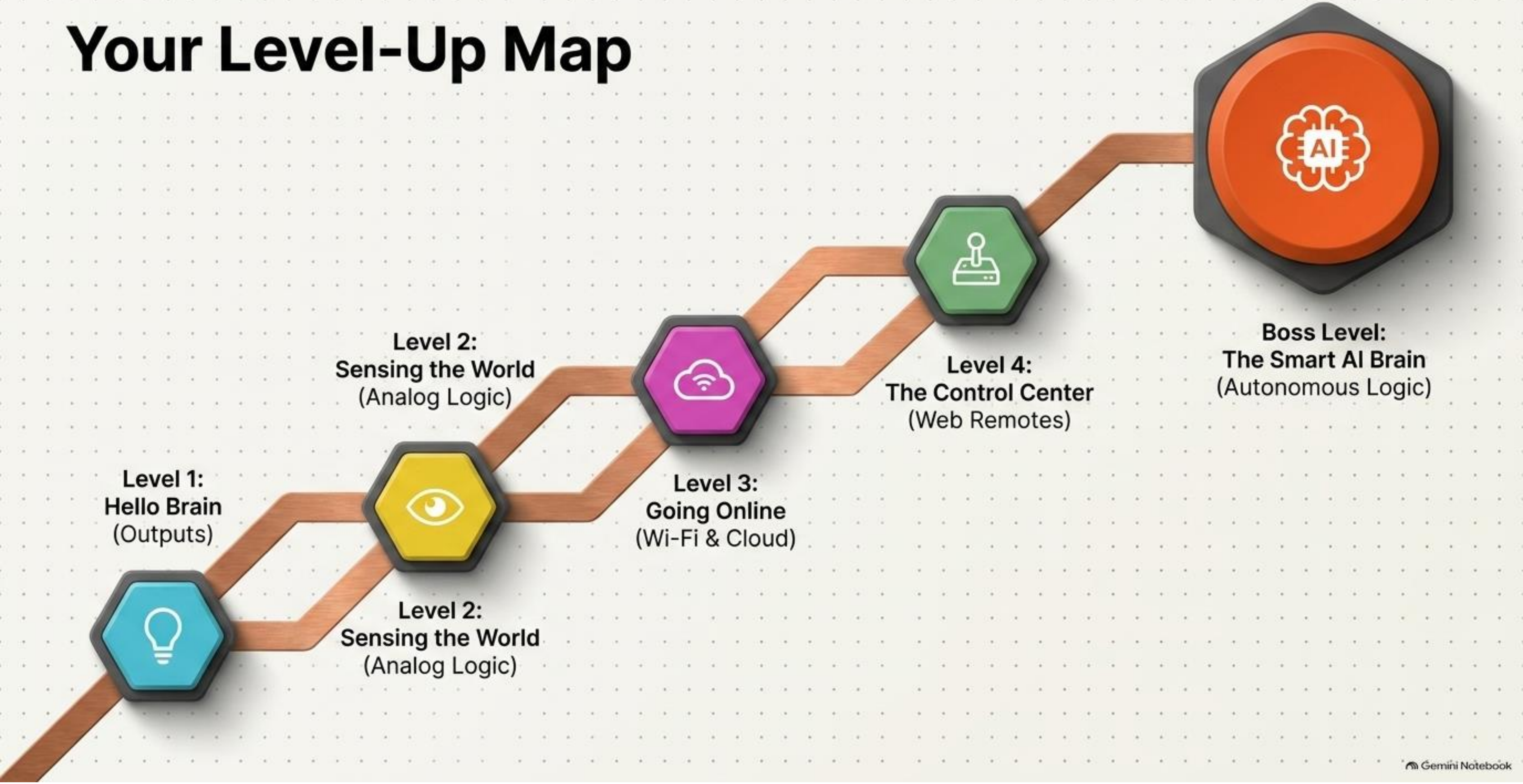
Your free MQTT web dashboard to store and visualize data from anywhere.



The IoT Master Loop

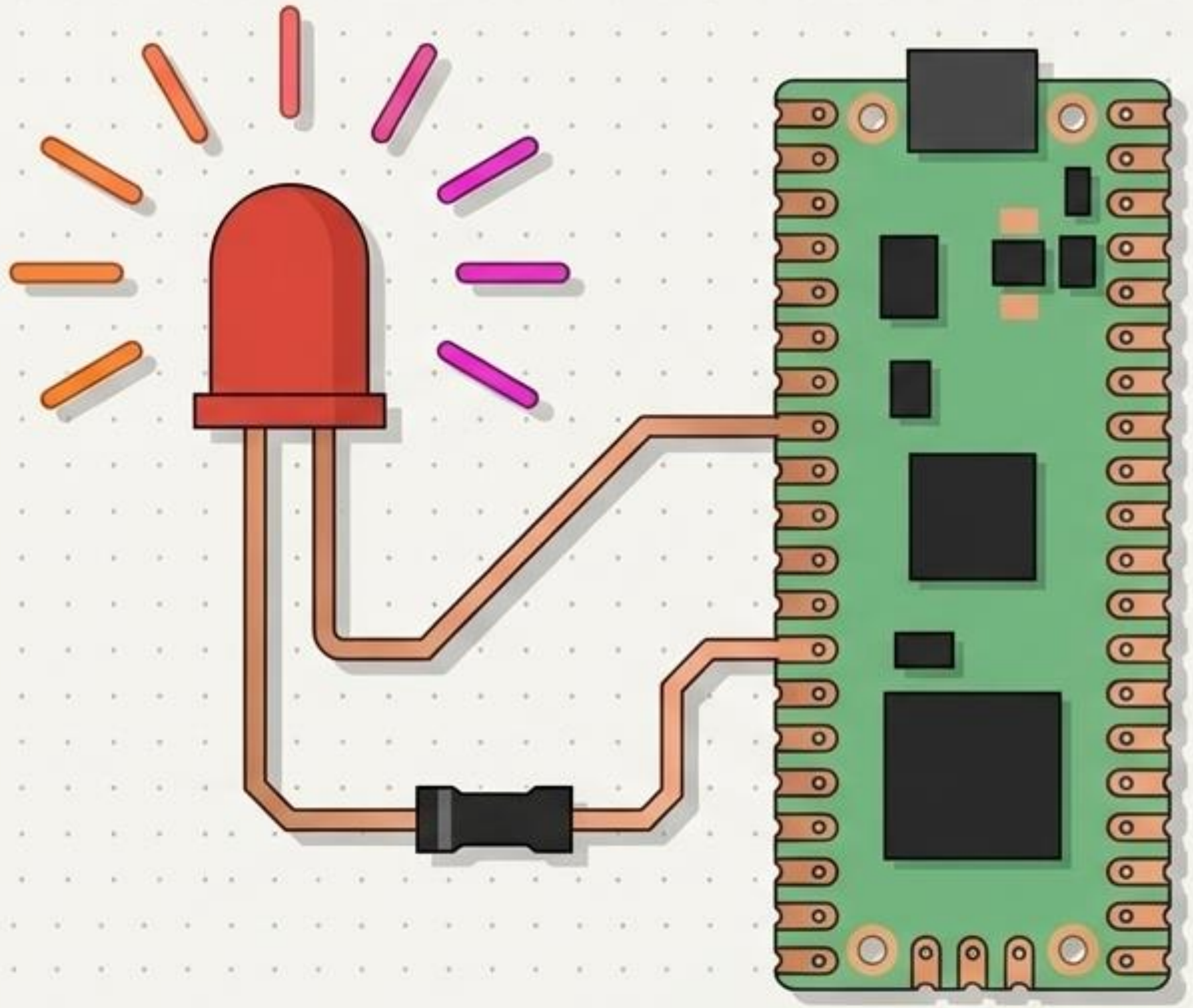


Your Level-Up Map



Level 1 🧠 Hello Brain!

Wake up your cyborg and test its pulse.

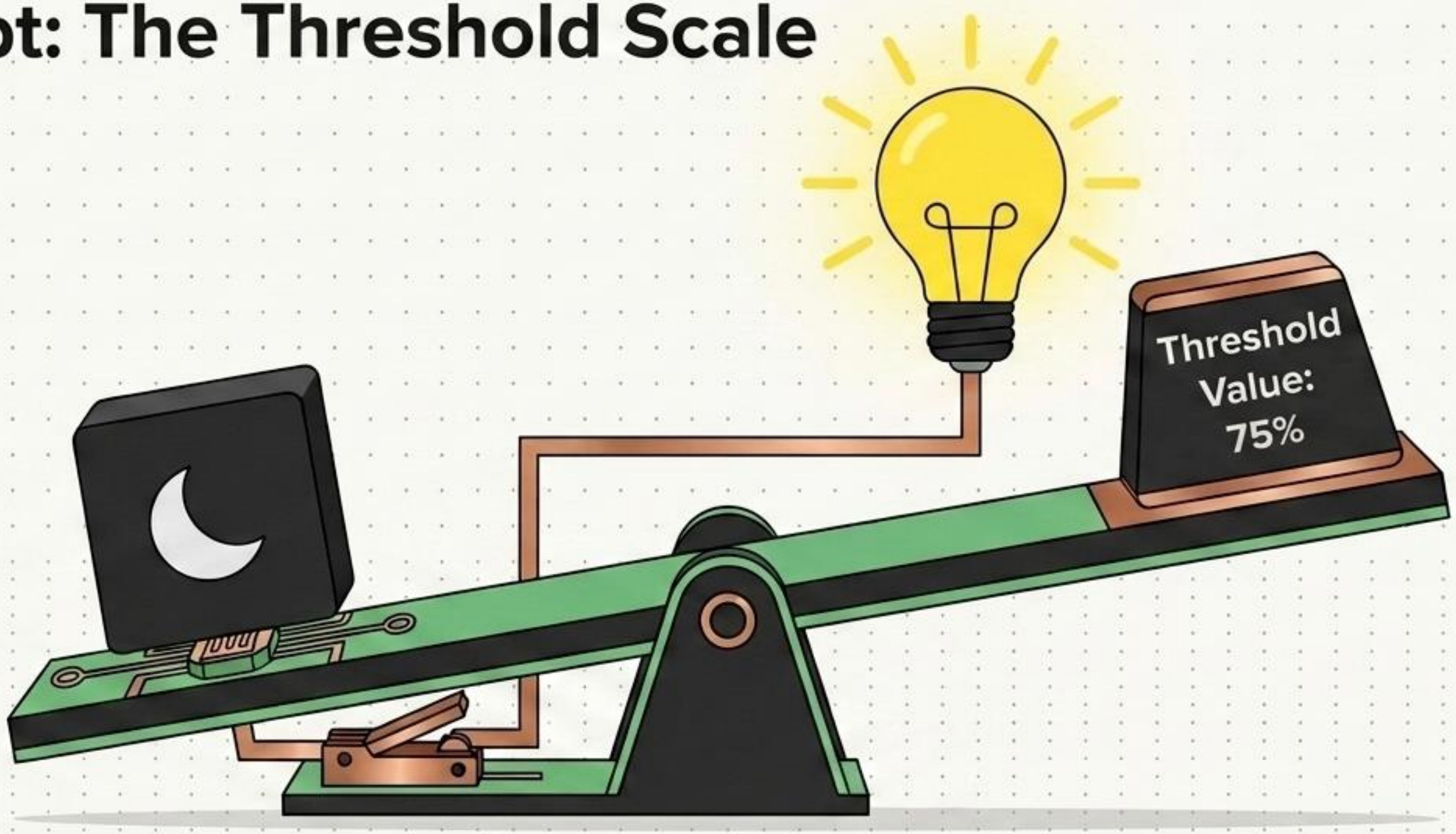


Hardware Unlocked:
LED + Resistor

Key Skill:
Basic GPIO Outputs

The Code:
Writing your very first timing loop to blink a light.

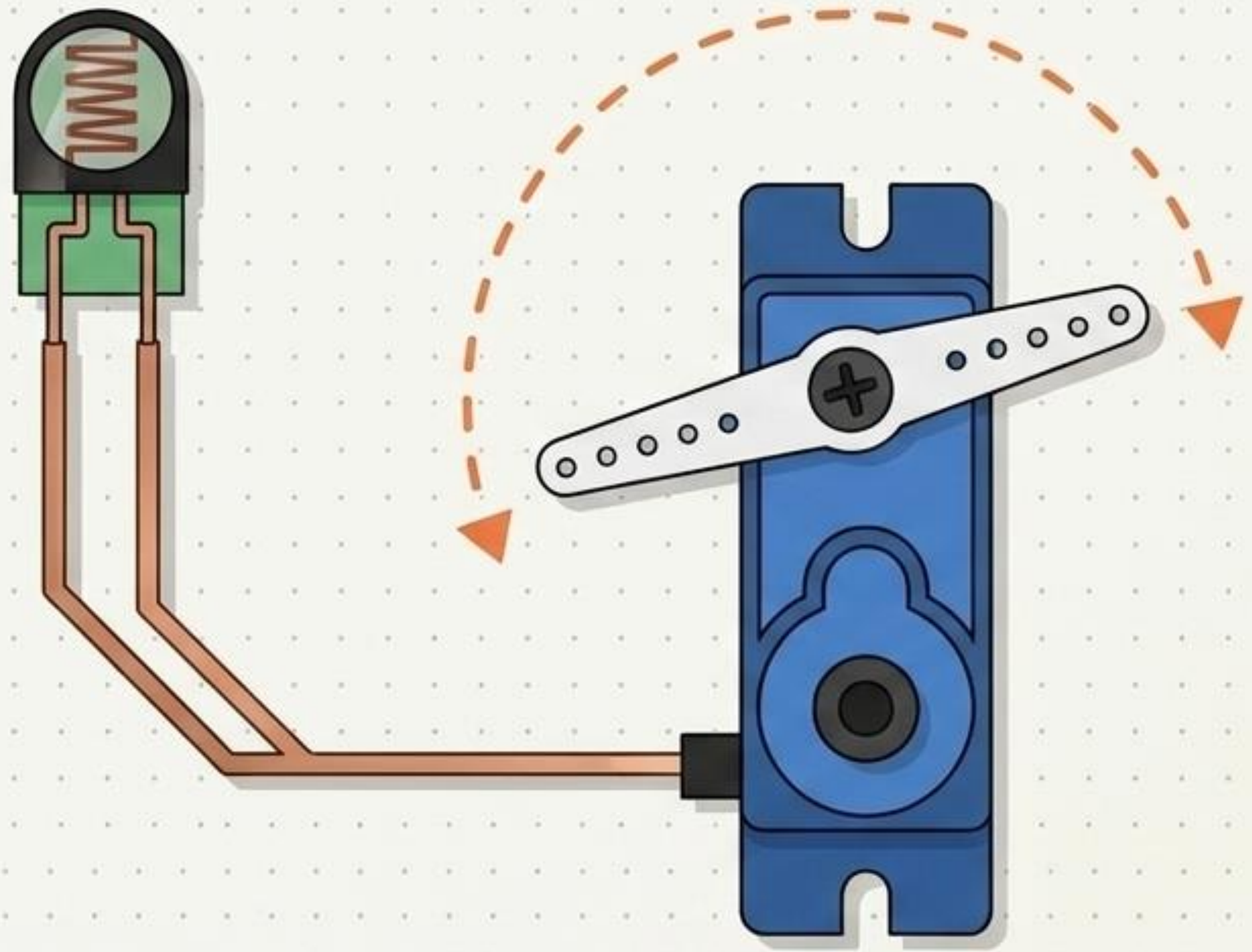
Concept: The Threshold Scale



How do we teach a machine when to act? We set a Threshold. If the room gets darker than our threshold line, the logic tips, closing the circuit, and the lights turn on automatically!

Level 2 ☀️ Sensing the World

Give your machine reflexes.

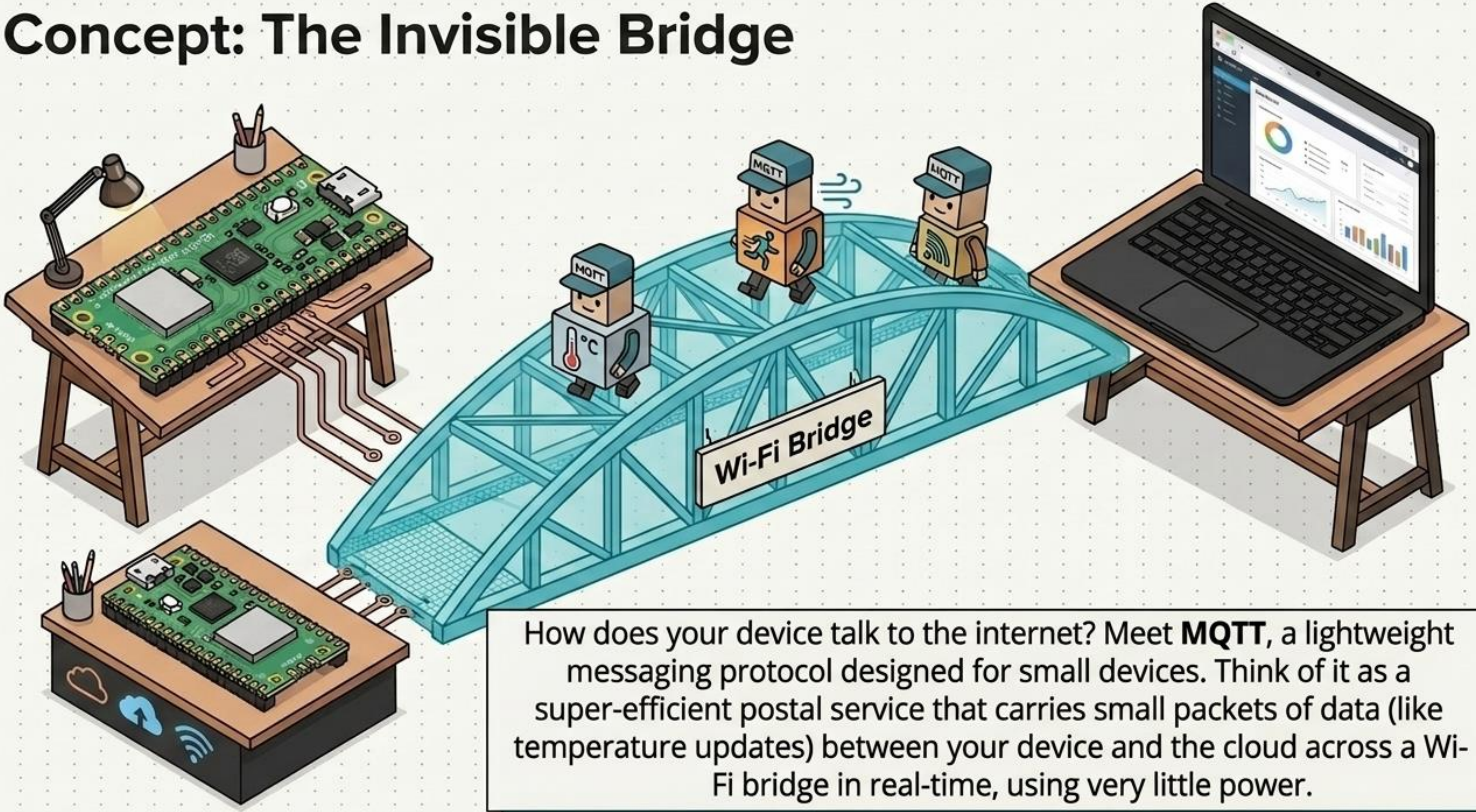


Hardware Unlocked:
LDR Sensor & SG90 Micro Servo

Key Skill:
Analog Inputs & Threshold Logic

```
The Action:
Program your device to move its
physical muscles (servo) based
on changes in ambient light.
```

Concept: The Invisible Bridge

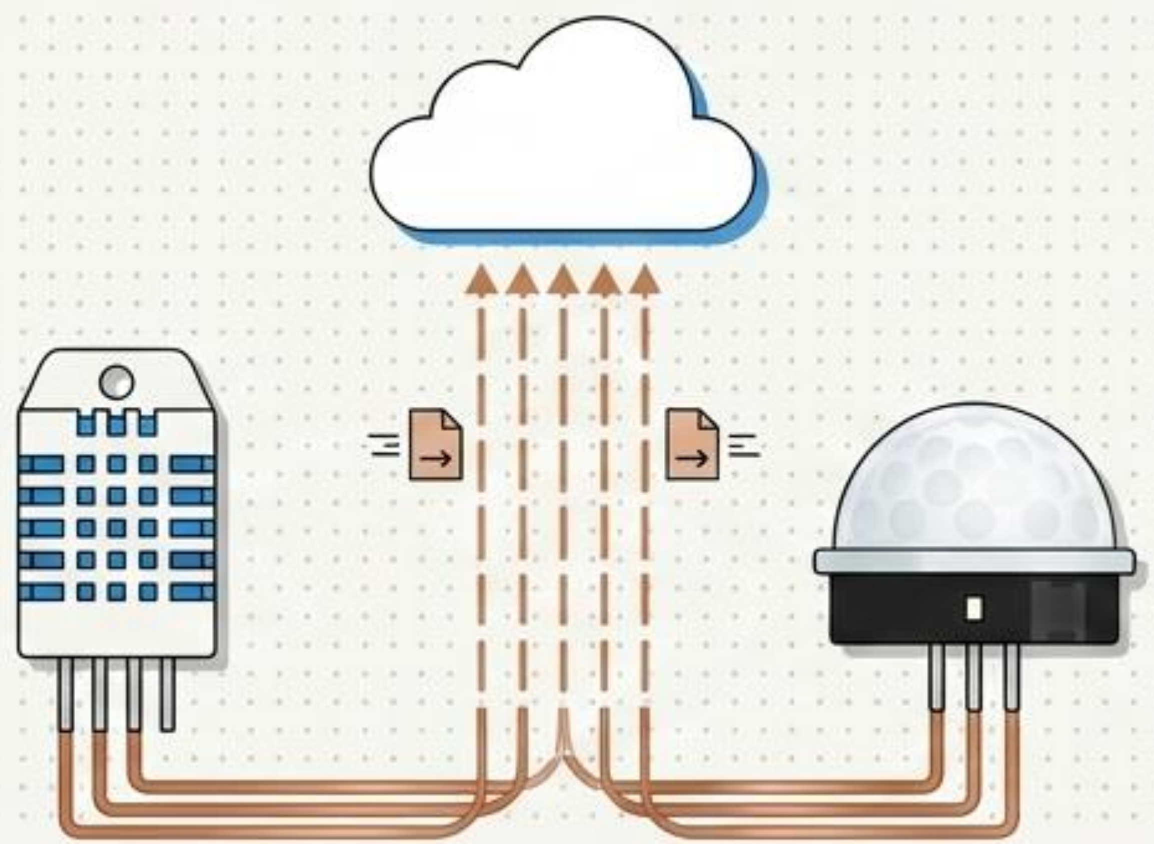


How does your device talk to the internet? Meet **MQTT**, a lightweight messaging protocol designed for small devices. Think of it as a super-efficient postal service that carries small packets of data (like temperature updates) between your device and the cloud across a Wi-Fi bridge in real-time, using very little power.

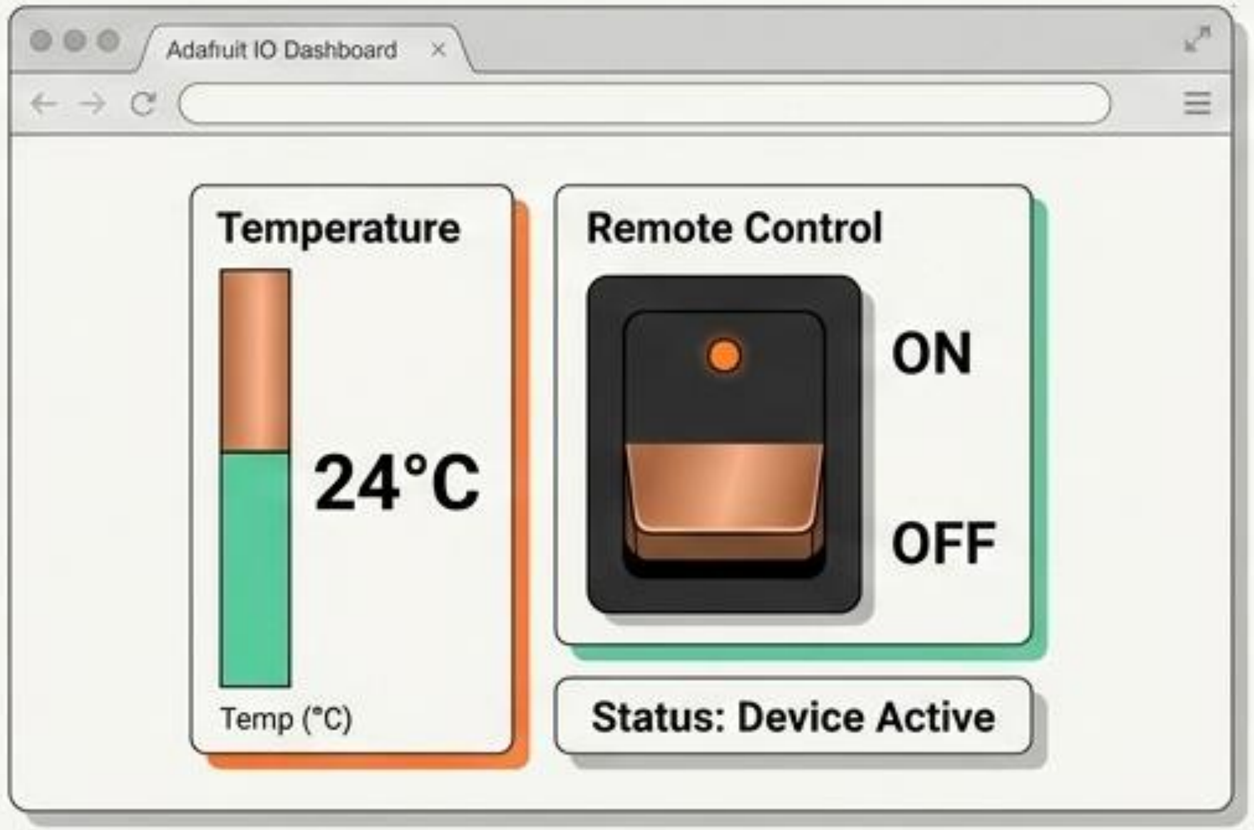
Levels 3 & 4 🛰️ Global Connection

Take your device online and build its control center.

Hardware & Uplink



The Dashboard



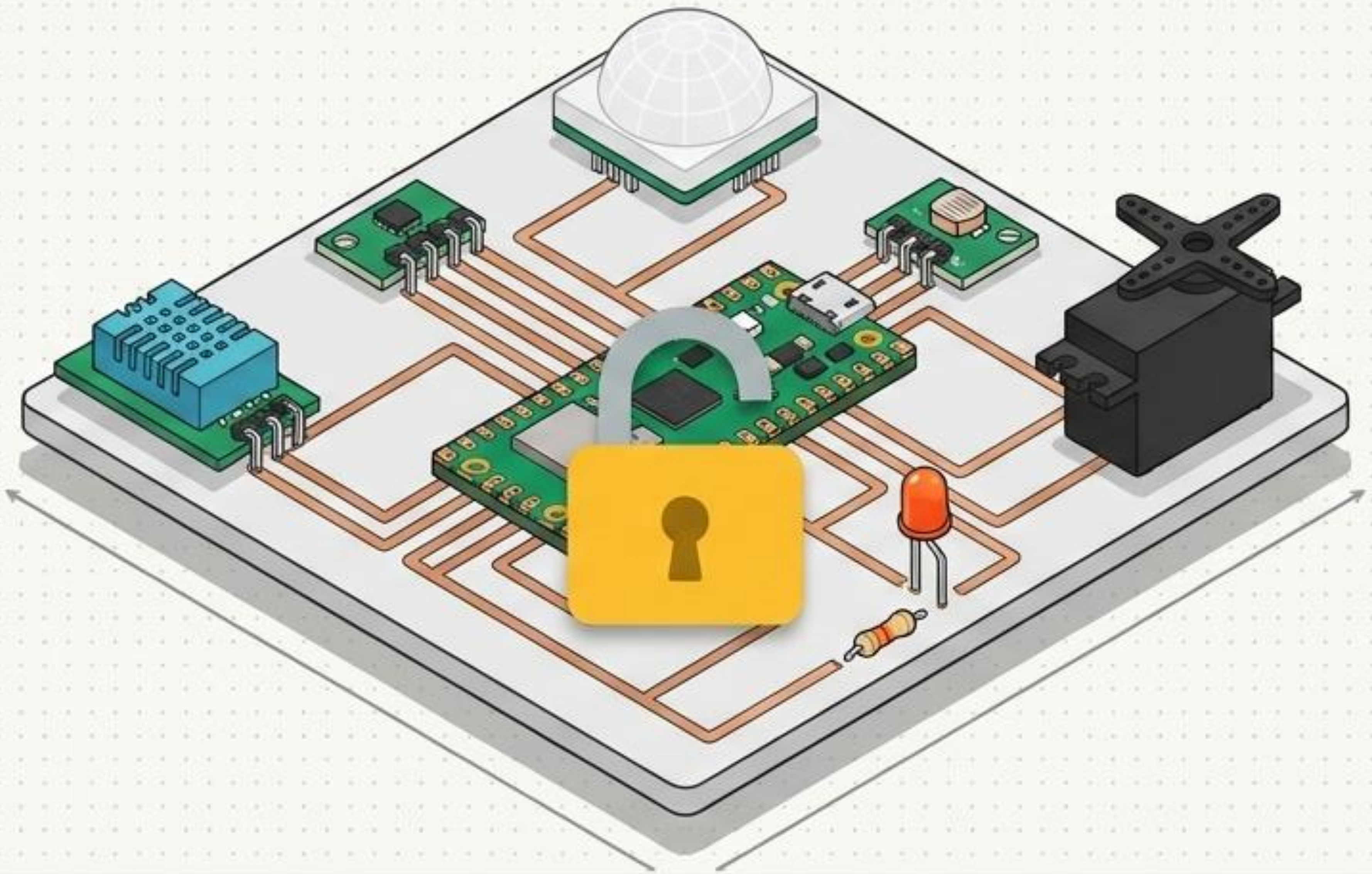
Hardware Unlocked:
`DHT22, PIR Motion Sensor, Wi-Fi`

Key Skill:
`MQTT Publishing, HTML, Web Visualization`

The Action:
Send real-time temperature and security data to your custom `Adafruit IO` web dashboard, and control your device remotely from your browser.

Boss Level 🤖 The Smart AI Brain

Bring it all together into a fully autonomous system.



Hardware Unlocked:
All Hardware Combined

Key Skill:
Rule-Based Edge AI

The Action:
No more manual control. Write the final rules that allow your machine to monitor its environment, make independent decisions, and manage your bedroom or farm autonomously!

Initiation Sequence

Ready to build? Complete your pre-flight checklist.

- ☑ **Check 1: Secure the Cloud** ☁️ – Create your free Adafruit IO Account. Save your Username & AIO Key!
- ☑ **Check 2: Prep the Tools** 💻 – Open Thonny IDE. Select "CircuitPython" as your interpreter.
- ☑ **Check 3: Power the Brain** ⚡ - Plug in your Raspberry Pi Pico W via Micro-USB.

▶ **INITIATE:** Watch the Getting Started Video & Dive into Level 1!